

COSC 4397 - Project Report

Jeffery Young
Student ID: 1022263

June 16, 2026

1 Problem

Given a query containing one aspect and one opinion, we want to provide the most relevant reviews (semantically and contextually) from the database. We want to go beyond keyword retrieval (semantic only) and get to opinion (contextual) retrieval.

2 Method

At least 2 methods will be used beyond a simple Boolean search (ie: N-gram based filtering, LDA, et cetera)

Each method used can be assessed by its precision which is the percentage found by dividing the number of relevant postings by the total number of postings retrieved.

- Boolean Search Boolean Search is a method about ..
- Method 1 Method 1 is a method about ...
- Method 2 Method 2 is a method about ...

3 Implementation

Python is used because of its native libraries.

- Step 0: Preprocessing - Filtering - Those appearing less than 5 times - Stops words from <https://gist.github.com/sebleier/554280> - Lemmatization/stemming - Smiley faces, (emojis?), and their variations using regular expressions, et cetera
- Step 1: Create postings list by Doc ID
- Step 2: Get Boolean Search up and running
- Step 3: Find a way to search filter the reviews that mention the aspect in the orientation that is provided in the opinion.

4 Results